

NFV and the Service Provider Cloud

Network Functions Virtualisation (NFV) involves virtualising entire network node functions into entities that may be connected or chained together, to create networking services.

This article examines the rise of Network Functions Virtualisation or NFV (which is built using cloud technologies in the service provider network), provides an overview of the ETSI reference architectural framework for NFV, and concludes with the role that open source projects play in this emerging area.

The service provider's dilemma

The demand for networks is increasing. Propelled by smartphones and social networking applications, people now produce and consume more pictures and videos every day resulting in an exponentially growing volume of traffic. With increasing network penetration, more and more people and devices access the

Internet every day. It is estimated that there are approximately 9 billion devices connected to the Internet already. While this may seem huge, technologists predict an even more rapid increase in Internet connected devices (to around 50 billion devices by 2020, according to some reports), a trend now called the Internet of Things (IoT).

While this appears to put network (Internet) service providers in a sweet spot, there is a problem. The revenues do not grow as fast, since more bandwidth and downloads need to be provided at lower costs. Handling the huge increase in scale (number of devices, number of flows) and throughput requires large capacity addition. They need to buy more networking equipment, wire it together, manage and operate it. Today's network design does not make this easy.

This results in a dilemma—not catering to the rising network